Notifications: How They Work and How to Maintain Them

Online Mindfulness Training Program

CITS3200 Group 45

1. What this document is for

This guide explains how reminders and notifications work in the program, so that someone who did not build the system can keep it running. You do not need to be a programmer to follow sections 1 to 6. Sections 7 to 10 are for whoever looks after the server or the code.

The short version: students get reminders as notifications on their phone or computer, the same way an app would notify them. The website sends them automatically. Once the server is set up, there is no daily work to do.

2. What students receive

There are five kinds of notification. Students control three of them from their Profile page under Notification Preferences.

Notification	When it is sent	Starts as	Student can switch off
Daily meditation reminder	At 8am on the day a meditation is scheduled. Not sent if they have already completed it.	On	Yes, in Profile
Daily survey reminder	At 8am each day while the student has an unlocked weekly survey they have not submitted.	Off	Yes, in Profile
Survey unlocked	When the student finishes all of a week's content and its survey becomes available.	Off	Yes, in Profile
New week available	At about 7am on the day a new week opens for the student's cohort. If it fails to send, it is retried each hour until 8pm.	On	No
Content updated	When an administrator publishes or edits a week the student can already open. At most once an hour per week.	On	No

Three rules apply to all five:

- **All times are Perth time.** Daily reminders go out at 8am. The hour is stored separately for each student, but there is no screen to change it yet, so it is 8am for everyone.

- **No student gets the same reminder twice.** Every send is recorded, and a reminder already sent is skipped.
- **A student who blocks notifications gets nothing.** The system respects a refusal and does not fall back to another channel.

> **Email is off.** The client asked for in-app reminders only. The code can send email as a backup, but it stays switched off as long as the email settings are left blank on the server (see section 7).

3. How it works

The system uses a standard, free feature built into every modern browser called Web Push. There is no paid messaging service and no account to renew.

When a student turns notifications on

1. The student opens their Profile page and clicks **Set up notifications**. A window shows instructions for their device.
2. They click **Enable notifications**, and their browser asks for permission. They click **Allow**.
3. The browser creates a private address for that device, called a subscription, and the website saves it against the student's account.

This is done once per device. A student who enables notifications on a laptop gets nothing on their phone until they enable it there too.

When a reminder goes out

1. Every hour, the server runs a small job that checks which students are due a reminder right now.
2. For each of those students, the server sends a short message to the saved address for each of their devices.
3. The browser maker's delivery service (Google, Apple, Mozilla or Microsoft) passes it to the device, even if the website is closed.
4. The device shows the notification. Clicking it opens the website at the right page.
5. The server records that the reminder was sent, so it is not sent again.

How the system decides who is due

A daily meditation reminder is sent to a student only when all of these are true:

- The student has an active enrolment in a cohort that has started and not yet ended.
- A meditation is scheduled for today. Each meditation has a **Reminder day** within its week, and days are counted from the cohort's start date.
- The current hour matches the student's reminder hour, which is 8am.
- The student has the daily content reminder switched on and has not blocked notifications.
- The student has not already completed that meditation or been reminded about it today.

4. What the administrator needs to do

Day to day, nothing. These four situations need action.

When new students join: ask them to turn notifications on

Notifications are off until each student turns them on, and the site does not ask them automatically. Include these steps in the welcome message, to be done the first time they log in:

1. Log in and open your **Profile** page.
2. Click **Set up notifications**. A window opens with instructions for your device.
3. Choose your device at the top: **iPhone / iPad**, **Android** or **Desktop**. The window picks the one it detects, and you can switch.
4. Follow the steps shown, click **Enable notifications**, then choose **Allow** when your device asks.
5. Repeat on every device you want reminders on. Each phone, tablet and computer is set up separately.

iPhone and iPad users have one extra step, shown in the window: they must add the site to their Home Screen first and open it from that icon.

Starting a new cohort

Go to **Admin > Cohorts** and create the cohort with the correct start and end dates. Every reminder date is worked out from the start date, so a wrong start date shifts all reminders. Then approve and enrol the students.

Adding or changing a meditation

Go to **Admin > Modules**, open the week, and add or edit the audio item. Set its **Reminder day**: Day 1 is the first day of that week, Day 2 the second, and so on. An item showing "No reminder" will never trigger one. Then publish the week.

> **Tip.** Set up and check content while the week is unpublished. Editing an unpublished week notifies nobody. Publishing sends one "content updated" notification to students who can already open that week.

Checking the system is healthy

The website has no status screen for notifications yet. The person who manages the server can run the three checks in section 8. A simple sign that something is wrong: several students say reminders have stopped on the same day.

5. Helping a student who gets no reminders

Work through these in order. Most problems are one of the first three.

1. **Have they enabled notifications on this device?** Ask them to open Profile, click Set up notifications, and click Enable notifications. It must be done on each device and each browser they use.

2. **Did they block the permission?** If the window says notifications are blocked, they need to allow them in their browser settings. The setup window explains how for each device.
3. **Is their device silencing the browser?** Do Not Disturb, Focus modes, and the device's own notification settings override the website.
4. **On iPhone or iPad, are they using the Home Screen app?** Reminders only work there after the site is added to the Home Screen and opened from that icon. It needs iOS 16.4 or later.
5. **Is the daily content reminder switched on in their Profile?**
6. **Have they already completed today's meditation?** No reminder is sent for completed items.

What each device supports

Device	Works in	What the student must do
Windows, Mac or Linux computer	Chrome, Edge, Firefox, Safari	Click Allow. The browser must be running to receive reminders.
Android phone or tablet	Chrome, Firefox, Edge, Samsung Internet	Tap Allow. Reminders arrive even with the browser closed.
iPhone or iPad, iOS 16.4 or later	Home Screen app only	Share, Add to Home Screen, open from the icon, then enable.
iPhone or iPad, older iOS	Not supported	Update iOS.

6. Four rules that must not be broken

Breaking any of these stops reminders for every student.

1. **Never replace the security keys.** The server holds a pair of keys (called VAPID keys) that every student's device is tied to. If they are regenerated or lost, every student has to enable notifications again. When the site moves to a new server, carry the same keys across. Keep a backup copy somewhere safe.
2. **Keep the site on HTTPS.** Browsers refuse to deliver notifications from a site without a valid security certificate. If the certificate expires, reminders stop.
3. **Keep the hourly job running.** Nothing is sent unless the scheduler runs the reminder commands each hour. It should restart by itself after a server reboot.
4. **Do not restart the server straight after publishing content.** The "content updated" notifications go out in the background for a short time after publishing. A restart during that time drops the rest, and they are not retried. Daily reminders are not affected.

7. Server setup (for IT or a developer)

Five things must be in place for notifications to work on a live server. On a developer's own computer the first two are not needed, which is why it works locally without them.

Requirement	Detail
HTTPS on a real domain	A certificate trusted by browsers. A self-signed certificate is not enough.
Outbound internet access	The server must be able to reach, on port 443: `fcm.googleapis.com` (Chrome, Android), `updates.push.services.mozilla.com` (Firefox), `*.push.apple.com` (Safari, iPhone), `*.notify.windows.com` (Edge). If the network uses a proxy, set HTTPS_PROXY for the backend.
VAPID keys	Generated once per environment and stored as secrets. Steps are in section 1 of the repository README.
An hourly scheduler	Runs the two commands in the next table. See Appendix A.
Service worker at the site root	The file `client/public/sw.js` is served by the frontend at `/sw.js`. Any reverse proxy in front must pass it through.

Settings (environment variables)

Setting	Value	Secret
`VAPID_PUBLIC_KEY`	The Application Server Key printed when the keys are generated.	No
`VAPID_PRIVATE_KEY`	The private key as a base64url text string, not a file.	Yes
`VAPID_ADMIN_EMAIL`	A real, monitored email address, written without `mailto:`. The default is a placeholder and must be replaced.	No
`ALLOWED_HOSTS`, `CORS_ALLOWED_ORIGINS`, `CSRF_TRUSTED_ORIGINS`	The site's own addresses, comma-separated. Without these the server rejects requests from the live site.	No
`NEXT_PUBLIC_API_BASE_URL`	The backend's address. It is fixed when the frontend is built, so it must be set before building.	No
`EMAIL_BACKEND` and other `EMAIL_` settings	Leave unset. This keeps the email backup switched off.	n/a

> **Watch for this.** If the VAPID settings are missing, nothing fails loudly. The server skips the push and writes only a warning to its log. Check these settings first whenever no reminders arrive.

The two scheduled commands

Command	How often	What it does
`python manage.py send_due_reminders`	Every hour	Sends daily meditation and survey reminders to students whose reminder hour is now.

Command	How often	What it does
`python manage.py send_module_unlock_notifications`	Every hour from 7am to 8pm Perth time	Tells students a new week is available on the day it opens. The 7am run sends the notices. Later runs only retry any that failed.

Both are safe to run more than once. A reminder that was sent is never sent again, and one that failed is retried on the next run.

8. Health checks and troubleshooting

Three commands for checking

Run these on the server. Under Docker, put `docker compose exec web` in front, replacing `web` with the name of the backend service.

Command	What it tells you
`python manage.py send_due_reminders --verify`	Lists meditations with no Reminder day, days with nothing scheduled, and duplicates. Run it before a cohort starts.
`python manage.py send_due_reminders --dry-run`	Shows who would be reminded right now, without sending anything.
`python manage.py push_adoption`	Shows how many enrolled students have notifications switched on.

Common problems

What you see	Likely cause	What to do
Nobody receives anything, no errors	Scheduler stopped, or VAPID settings empty	Check the scheduler is running, then check the VAPID settings.
Dry run lists nobody	No active enrolments, no meditation scheduled today, or it is not the students' reminder hour	Check cohort dates, enrolments and Reminder days.
Send command times out	Firewall blocks the delivery services	Ask IT to allow the hosts in section 7.
Worked, then stopped for everyone after a server change	VAPID keys were replaced	Restore the original keys from backup.
Reminders arrive at the wrong hour	Server clock or time zone setting	The code uses Perth time. Check the server's clock is correct.

What you see	Likely cause	What to do
No permission prompt appears for a student	They blocked it earlier, or already allowed it	If blocked, allow it in browser settings. If already allowed, it is working.
iPhone says notifications are not supported	Site opened in Safari, not from the Home Screen icon	Add to Home Screen and open from the icon.

9. Where the code lives (for a developer)

File	What it does
`client/src/components/notifications/NotificationManager.tsx`	Asks for permission, creates the subscription and saves it to the backend.
`client/src/components/notifications/NotificationSetupModal.tsx`	The setup window opened from the Profile page.
`client/src/components/notifications/notificationSetupContent.ts`	The instruction text for each device. Edit this to change the wording.
`client/public/sw.js`	Service worker. Shows each notification and opens the site when it is clicked.
`client/public/manifest.json`	Lets the site be installed to a phone's Home Screen. Needed for iPhone.
`app/notifications.py`	Sends a push, removes expired subscriptions, and holds the content-update and survey-unlocked logic.
`app/management/commands/send_due_reminders.py`	The hourly reminder command, including `--verify` and `--dry-run`.
`app/management/commands/send_module_unlock_notifications.py`	The "new week available" command.
`app/management/commands/push_adoption.py`	The adoption report.
`app/signals.py`	Triggers the content-update notification when a week or item is saved.
`app/models.py`	`PushSubscription`, `NotificationPreference` and `NotificationLog`.
`server/settings.py`	Reads the settings listed in section 7.

The notification log (`NotificationLog`) records every send with its type, channel, time and whether it succeeded. It is visible in Django admin and is the first place to look when checking what a student was sent.

10. Status at handover

Tested and working on a developer's computer (3 October 2026)

- Enabling notifications and saving the subscription.
- Delivering a push from the server to the browser, in Safari on a Mac.
- The reminder command choosing the right student, at their reminder hour, for the meditation due that day.
- A second run on the same day sending nothing.

The Chrome, Android and iPhone paths use the same code but were not run as part of this check.

Not yet tested

These need the hosting environment from UWA IT, which was not available when this was written.

- Delivery over HTTPS on the real domain.
- Delivery on an iPhone from the Home Screen app.
- The hourly scheduler running unattended.

Known gaps

- **Reminder hour does not match the client's request.** At client meeting 3 (11 September 2026) Sarah asked for the daily reminder to go out at 5 to 6pm if the meditation has not been done. The Profile page says "around 5pm", but the system currently sends at 8am. The default hour needs changing to 5pm.
- **Survey-unlocked notification starts switched off.** At the same meeting Sarah said students should be notified when their weekly survey unlocks. Students currently have to switch this on themselves in their Profile.
- **No screen to change the reminder hour.** It is 8am for everyone unless changed in Django admin.
- **Docker setup.** The Docker files, including the scheduler in Appendix A, were still being written.
- **Home Screen icon.** The app icons are placeholders until the final logo is supplied.

Appendix A. Scheduler setup

UWA IT deploys with Docker Compose, so the simplest scheduler is one extra service in the Compose file. It reuses the backend image and runs both commands on the hour. Match the build path, volume and service names to the final Compose file.

```
  scheduler:
    build: .
    env_file: .env
    volumes:
      - db-data:/data
    restart: unless-stopped
    command: >
      sh -c 'while true; do
```

```
      sleep $$((3600 - $$(date +%s) % 3600));
      python manage.py send_due_reminders;
      h=$$(( ($$(date +%s) / 3600 + 8) % 24 ));
      if [ $$h -ge 7 ] && [ $$h -le 20 ]; then
      python manage.py send_module_unlock_notifications;
      fi;
      done'
```

The scheduler must use the same database as the backend service. The `h` line works out the hour in Perth, so the "new week" notices only go out between 7am and 8pm.

If IT prefers to schedule on the server itself, these two cron lines do the same job. Replace `web` with the backend service name. Each entry must be typed on a single line, even where it wraps here.

```
0 * * * * cd <app folder> && docker compose exec -T web python manage.py
send_due_reminders

5 23,0-12 * * * cd <app folder> && docker compose exec -T web python manage.py
send_module_unlock_notifications
```

The second line's hours are in UTC and equal 7am to 8pm in Perth.

Appendix B. Terms used

Term	Meaning
Push notification	A message that pops up on a phone or computer, even when the website is not open.
Subscription	The private address a browser creates for one device so the server can send it notifications.
Service worker	A small script the website installs in the browser. It receives each notification and displays it.
VAPID keys	A pair of security keys that prove a notification really came from this website.
Scheduler	The background job that runs the reminder commands every hour.
Cohort	A group of students doing the program over the same dates.
Reminder day	The day within a week on which a meditation's reminder is sent. Day 1 is the first day of that week.
Environment variable	A setting stored on the server outside the code, used for things like keys and addresses.